



UNIVERSITÄT  
LEIPZIG

Tutorial

# Natural Language Processing – Week 1

Jacob Suchardt

# AGENDA

- Introduction
- Exercise sheet

# INTRODUCTION

Your tutor(s):

Jacob Suchardt (jacob.suchardt@uni-leipzig.de)

[ ja:kɔp    zuχa:t ]

- PhD Student Natural Language Processing @ Leipzig University & ScaDS.AI
- Background: M.A. Language Technology
  - And before that: (computational linguistics & Japanese language and literature)

## Tutorials

- Optional, as we will mainly be discussing the optional exercise sheets
- Wednesdays: 11-13 (Toshiki), 15-17 (Jacob), SG 3-12
- Use same Moodle course as the lecture

# EXERCISE SHEET

We will discuss these in the tutorials  
But usually there won't be 'model solutions'



## Natural Language Processing

### Exercise Sheet 1: NLP Problems

#### Exercise 1

Think of an application of natural language processing that did **not** appear in the lecture slides and briefly describe it. Ideally, choose an application that your fellow students are unlikely to have thought of.

#### Exercise 2

For the paragraph below, referring to the examples shown in the lecture slides, do the following **by hand**:

- a) Tokenisation
- b) Lemmatisation
- c) POS tagging
- d) Named Entity Recognition
- e) Identify all coreferences
- f) Entity relation extraction
- g) Count unigrams, bigrams, and trigrams and create a frequency list of each

What were challenging cases where you had to make a decision, and how did you make it?

#### Paragraph:

The wise Sir Bedemir was the first to join King Arthur's knights, but other illustrious names were soon to follow: Sir Lancelot the Brave; Sir Galahad the Pure; and Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of Angnor, who had nearly stood up to the vicious Chicken of Bristol and who had personally wet himself at the Battle of Badon Hill; and the aptly named Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the Knights of the Round Table.



## Exercise 1

# EXERCISE SHEET

Think of an application of natural language processing that did **not** appear in the lecture slides and briefly describe it. Ideally, choose an application that your fellow students are unlikely to have thought of.

- 1. Aid humans in writing.

Correcting mistakes, formulating and paraphrasing text, transcription.

- 2. Identify texts related to spoken or written requests.

Text information retrieval, semantic text similarity, question answering.

- 3. Make sense of texts without reading the originals.

Categorization, information extraction, summarization, translation.

- 4. Instruct and be advised by a computer.

Audio interfaces (e.g., dialog systems, robotics), learning and assessment.

- 5. Converse with computers as if they were human.

Turing test, Conversational AI and chatbots, argumentation, affective computing.

- 6. Understanding the nature of language and its relation to (artificial) intelligence.

Pragmatics, causality, alignment and grounding, multimodality, simulation, linguistics.

Leonie Weissweiler, slides adapted from

Hagen/Pothast/Stein/Wachsmuth/Weigmann/Wolska

09.04.2026

## Your ideas?

- Plagiarism & authorship detection
- Screen readers

Mine: Image captioning & guides for visually impaired individuals (Baghel et. al, 2025)

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following **by hand**:

- a) Tokenisation
- b) Lemmatisation
- c) POS tagging
- d) Named Entity Recognition
- e) Identify all coreferences
- f) Entity relation extraction
- g) Count unigrams, bigrams, and trigrams and create a frequency list of each

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

a) Tokenisation

or: Arthur | ' | s

The|wise|Sir Bedemir|was|the|first|to|join|King Arthur's|knights|,|but|other|illustrious|names|were|soon|to|follow|:| Sir Lancelot the Brave|;| Sir Galahad the Pure|;| and| Sir Robin the Not-quite-so-brave-as-Sir-Lancelot |who| had| nearly| fought| the| Dragon of Angnor|,| who| had| nearly| stood| up| to| the| vicious| Chicken of Bristol| and| who| had| personally| wet| himself| at| the| Battle of Badon Hill|;| and| the| aptly| named| Sir Not-appearing-in-this-film|.| Together| they| formed| a| band| whose| names| and| deeds| were| to| be| retold| throughout| the| centuries|,| the| Knights of the Round Table|.

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

a) Tokenisation

The|wise|Sir|Bedemir|was|the|first|to|join|King|Arthur's|knights|,|but|other|illustrious|names|were|soon|to|follow|;|Sir|Lancelot|the|Brave|;|Sir|Galahad|the|Pure|;|and|Sir|Robin|the|Not-quite-so-brave-as-Sir-Lancelot|who|had|nearly|fought|the|Dragon|of|Angnor|,|who|had|nearly|stood|up|to|the|vicious|Chicken|of|Bristol|and|who|had|personally|wet|himsel|f|at|the|Battle|of|Badon|Hill|;|and|the|aptly|named|Sir|Not-appearing-in-this-film|. Together|they|formed|a|band|whose|names|and|deeds|were|to|be|retold|throughout|the|centuries|,|the|Knights|of|the|Round|Table|.

Naïve tokenization without accounting for multi-word proper nouns

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

b) Lemmatisation

the King Arthur knight  
The wise Sir Bedemir was the first to join King Arthur's knights, but other illustrious  
name be  
names were soon to follow: Sir Lancelot the Brave; Sir Galahad the Pure; and Sir  
have fight  
Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of  
have stand have  
Angnor, who had nearly stood up to the vicious Chicken of Bristol and who had  
name  
personally wet himself at the Battle of Badon Hill; and the aptly named Sir Not-  
together form name deed be  
appearing-in-this-film. Together they formed a band whose names and deeds were  
retell century  
to be retold throughout the centuries, the Knights of the Round Table.

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

c) POS tagging

## Universal Part-of-Speech (UPOS)

<https://universaldependencies.org/u/pos/>

### Open-class words

ADJ: adjective

ADV: adverb

INTJ: interjection

NOUN: noun

PROPN: proper noun

VERB: verb

### Closed-class words

ADP: adposition

AUX: auxiliary

CCONJ: coordinating conjunction

DET: determiner

NUM: numeral

PART: particle

PRON: pronoun

SCONJ: subordinating  
conjunction

### Other

PUNCT: punctuation

SYM: symbol

X: other

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

c) POS tagging

DET ADJ PROPON AUX DET ADJ PART VERB PROPON NOUN PUNCT CCONJ ADV ADJ  
The wise Sir Bedemir was the first to join King Arthur's knights , but other illustrious

NOUN AUX ADJ PART VERB PUNCT PROPON PUNCT PROPON PUNCT CCONJ PROPON  
names were soon to follow : Sir Lancelot the Brave ; Sir Galahad the Pure ; and Sir

-PROPON (cont.) PRON AUX ADV VERB DET PROPON  
Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of

-PROPON (cont.) PUNCT PRON AUX ADV VERB ADP PART DET ADJ PROPON CCONJ PRON AUX  
Angnor , who had nearly stood up to the vicious Chicken of Bristol and who had

ADV VERB PRON ADP DET PROPON PUNCT CCONJ DET ADV VERB PROPON  
personally wet himself at the Battle of Badon Hill ; and the aptly named Sir Not-

-PROPON (cont.) PUNCT ADV PRON VERB DET NOUN PRON NOUN CCONJ NOUN AUX  
appearing-in-this-film . Together they formed a band whose names and deeds were

PART AUX VERB ADP DET NOUN PUNCT DET PROPON PUNCT  
to be retold throughout the centuries , the Knights of the Round Table .

## Exercise 2

## EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

c) POS tagging

The wise Sir Bedemir was the first to join King Arthur's knights , but other illustrious  
 names were soon to follow : Sir Lancelot the Brave ; Sir Galahad the Pure ; and Sir  
 Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of  
 Angnor , who had nearly stood up to the vicious Chicken of Bristol and who had  
 personally wet himself at the Battle of Badon Hill ; and the aptly named Sir Not-  
 appearing-in-this-film . Together they formed a band whose names and deeds were  
 to be retold throughout the centuries , the Knights of the Round Table .



## Exercise 2

## EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

c) POS tagging

## Penn Treebank POS tagset

[https://web.stanford.edu/~jurafsky/slp3/old\\_oct19/8.pdf](https://web.stanford.edu/~jurafsky/slp3/old_oct19/8.pdf)

| Tag  | Description                   | Example             | Tag   | Description           | Example            | Tag  | Description          | Example              |
|------|-------------------------------|---------------------|-------|-----------------------|--------------------|------|----------------------|----------------------|
| CC   | coordinating conjunction      | <i>and, but, or</i> | PDT   | predeterminer         | <i>all, both</i>   | VBP  | verb non-3sg present | <i>eat</i>           |
| CD   | cardinal number               | <i>one, two</i>     | POS   | possessive ending     | <i>'s</i>          | VBZ  | verb 3sg pres        | <i>eats</i>          |
| DT   | determiner                    | <i>a, the</i>       | PRP   | personal pronoun      | <i>I, you, he</i>  | WDT  | wh-determ.           | <i>which, that</i>   |
| EX   | existential 'there'           | <i>there</i>        | PRP\$ | possess. pronoun      | <i>your, one's</i> | WP   | wh-pronoun           | <i>what, who</i>     |
| FW   | foreign word                  | <i>mea culpa</i>    | RB    | adverb                | <i>quickly</i>     | WP\$ | wh-possess.          | <i>whose</i>         |
| IN   | preposition/<br>subordin-conj | <i>of, in, by</i>   | RBR   | comparative<br>adverb | <i>faster</i>      | WRB  | wh-adverb            | <i>how, where</i>    |
| JJ   | adjective                     | <i>yellow</i>       | RBS   | superlatv. adverb     | <i>fastest</i>     | \$   | dollar sign          | <i>\$</i>            |
| JJR  | comparative adj               | <i>bigger</i>       | RP    | particle              | <i>up, off</i>     | #    | pound sign           | <i>#</i>             |
| JJS  | superlative adj               | <i>wildest</i>      | SYM   | symbol                | <i>+, %, &amp;</i> | "    | left quote           | <i>' or "</i>        |
| LS   | list item marker              | <i>1, 2, One</i>    | TO    | "to"                  | <i>to</i>          | "    | right quote          | <i>' or "</i>        |
| MD   | modal                         | <i>can, should</i>  | UH    | interjection          | <i>ah, oops</i>    | (    | left paren           | <i>[, (, {, &lt;</i> |
| NN   | sing or mass noun             | <i>llama</i>        | VB    | verb base form        | <i>eat</i>         | )    | right paren          | <i>], ), }, &gt;</i> |
| NNS  | noun, plural                  | <i>llamas</i>       | VBD   | verb past tense       | <i>ate</i>         | ,    | comma                | <i>,</i>             |
| NNP  | proper noun, sing.            | <i>IBM</i>          | VBG   | verb gerund           | <i>eating</i>      | .    | sent-end punc        | <i>. ! ?</i>         |
| NNPS | proper noun, plu.             | <i>Carolinas</i>    | VBN   | verb past part.       | <i>eaten</i>       | :    | sent-mid punc        | <i>: ; ... - -</i>   |

**Figure 8.1** Penn Treebank part-of-speech tags (including punctuation).

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

c) POS tagging

The wise Sir Bedemir was the first to join King Arthur's knights, but other illustrious names were soon to follow : Sir Lancelot the Brave ; Sir Galahad the Pure ; and Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of Angnor , who had nearly stood up to the vicious Chicken of Bristol and who had personally wet himself at the Battle of Badon Hill ; and the aptly named Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries , the Knights of the Round Table .

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

d) Named Entity Recognition

The wise PERSON Sir Bedemir was the first to join PERSON King Arthur's knights, but other illustrious names were soon to follow: PERSON Sir Lancelot the Brave; PERSON Sir Galahad the Pure; and Sir PERSON Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the PERSON Dragon of Angnor, who had nearly stood up to the vicious PERSON Chicken of Bristol and who had personally wet himself at the EVENT Battle of Badon Hill; and the aptly named Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the ORGANIZATION Knights of the Round Table.

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

e) Identify all coreferences

The wise Sir Bedemir was the first to join King Arthur's knights, but other illustrious names were soon to follow: Sir Lancelot the Brave; Sir Galahad the Pure; and Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of Angnor, who had nearly stood up to the vicious Chicken of Bristol and who had personally wet himself at the Battle of Badon Hill; and the aptly named Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the Knights of the Round Table.

## Exercise 2

## EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

f) Entity relation extraction

The wise PERSON Sir Bedemir was the first to join PERSON King Arthur's knights, but other illustrious names were soon to follow: PERSON Sir Lancelot the Brave; PERSON Sir Galahad the Pure; and PERSON Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the Dragon of Angnor, who had nearly stood up to the vicious Chicken of Bristol and who had personally wet himself at the EVENT Battle of Badon Hill; and the aptly named PERSON Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the ORGANIZATION Knights of the Round Table.

Member-of(PERSON, Knights of the Round Table)

## Exercise 2

## EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

f) Entity relation extraction

The wise PERSON Sir Bedemir was the first to join PERSON King Arthur's knights, but other illustrious names were soon to follow: PERSON Sir Lancelot the Brave, PERSON Sir Galahad the Pure, and PERSON - Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the PERSON - Dragon of Angnor, who had nearly stood up to the vicious PERSON Chicken of Bristol and who had personally wet himself at the EVENT Battle of Badon Hill, and the aptly named PERSON - Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the ORGANIZATION Knights of the Round Table.

## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

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The wise PERSON Sir Bedemir was the first to join PERSON King Arthur's knights, but other illustrious names were soon to follow: PERSON Sir Lancelot the Brave, PERSON Sir Galahad the Pure, and PERSON - Sir Robin the Not-quite-so-brave-as-Sir-Lancelot who had nearly fought the PERSON - Dragon of Angnor, who had nearly stood up to the vicious PERSON Chicken of Bristol and who had personally wet himself at the EVENT Battle of Badon Hill; and the aptly named PERSON - Sir Not-appearing-in-this-film. Together they formed a band whose names and deeds were to be retold throughout the centuries, the ORGANIZATION Knights of the Round Table.



## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following **by hand**:

g) Count unigrams, bigrams, and trigrams and create a frequency list of each

the 7  
to 4  
and 4  
, 3  
; 3  
who 3  
had 3  
names 2  
were 2  
nearly 2  
. 2  
The 1  
wise 1  
Sir Bedemir 1  
was 1  
first 1  
join 1  
King Arthur 1  
's 1  
knights 1

but 1  
other 1  
illustrious 1  
soon 1  
follow 1  
: 1  
Sir Lancelot the Brave 1  
Sir Galahad the Pure 1  
Sir Robin the Not-quite-so-brave-as-Sir-  
Lancelot 1  
fought 1  
Dragon of Angnor 1  
stood 1  
up 1  
vicious 1  
Chicken of Bristol 1  
personally 1  
wet 1  
himself 1  
at 1

Battle of Badon Hill 1  
aptly 1  
named 1  
Sir Not-appearing-in-this-film 1  
Together 1  
they 1  
formed 1  
a 1  
band 1  
whose 1  
deeds 1  
be 1  
retold 1  
throughout 1  
centuries 1  
Knights of the Round Table 1



## Exercise 2

# EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

g) Count unigrams, bigrams, and trigrams and create a frequency list of each

|                            |                                                               |                                               |                                         |
|----------------------------|---------------------------------------------------------------|-----------------------------------------------|-----------------------------------------|
| ('who', 'had') 3           | ('follow', ':') 1                                             | ('had', 'personally') 1                       | ('deeds', 'were') 1                     |
| (';', 'and') 2             | (';', 'Sir Lancelot the Brave') 1                             | ('personally', 'wet') 1                       | ('were', 'to') 1                        |
| ('had', 'nearly') 2        | ('Sir Lancelot the Brave', ';') 1                             | ('wet', 'himself') 1                          | ('to', 'be') 1                          |
| ('The', 'wise') 1          | (';', 'Sir Galahad the Pure') 1                               | ('himself', 'at') 1                           | ('be', 'retold') 1                      |
| ('wise', 'Sir Bedemir') 1  | ('Sir Galahad the Pure', ';') 1                               | ('at', 'the') 1                               | ('retold', 'throughout') 1              |
| ('Sir Bedemir', 'was') 1   | ('and', 'Sir Robin the Not-quite-so-brave-as-Sir-Lancelot') 1 | ('the', 'Battle of Badon Hill') 1             | ('throughout', 'the') 1                 |
| ('was', 'the') 1           | ('Sir Robin the Not-quite-so-brave-as-Sir-Lancelot', 'who') 1 | ('Battle of Badon Hill', ';') 1               | ('the', 'centuries') 1                  |
| ('the', 'first') 1         | ('nearly', 'fought') 1                                        | ('and', 'the') 1                              | ('centuries', ';') 1                    |
| ('first', 'to') 1          | ('fought', 'the') 1                                           | ('the', 'aptly') 1                            | (';', 'the') 1                          |
| ('to', 'join') 1           | ('the', 'Dragon of Angnor') 1                                 | ('aptly', 'named') 1                          | ('the', 'Knights of the Round Table') 1 |
| ('join', 'King Arthur') 1  | ('Dragon of Angnor', ';') 1                                   | ('named', 'Sir Not-appearing-in-this-film') 1 | ('Knights of the Round Table', ';') 1   |
| ('King Arthur', 's') 1     | (';', 'who') 1                                                | ('Sir Not-appearing-in-this-film', ';') 1     |                                         |
| ('s', 'knights') 1         | ('nearly', 'stood') 1                                         | (';', 'Together') 1                           |                                         |
| ('knights', ',') 1         | ('stood', 'up') 1                                             | ('Together', 'they') 1                        |                                         |
| (';', 'but') 1             | ('up', 'to') 1                                                | ('they', 'formed') 1                          |                                         |
| ('but', 'other') 1         | ('to', 'the') 1                                               | ('formed', 'a') 1                             |                                         |
| ('other', 'illustrious') 1 | ('the', 'vicious') 1                                          | ('a', 'band') 1                               |                                         |
| ('illustrious', 'names') 1 | ('vicious', 'Chicken of Bristol') 1                           | ('band', 'whose') 1                           |                                         |
| ('names', 'were') 1        | ('Chicken of Bristol', 'and') 1                               | ('whose', 'names') 1                          |                                         |
| ('were', 'soon') 1         | ('and', 'who') 1                                              | ('names', 'and') 1                            |                                         |
| ('soon', 'to') 1           |                                                               | ('and', 'deeds') 1                            |                                         |
| ('to', 'follow') 1         |                                                               |                                               |                                         |

## Exercise 2

### EXERCISE SHEET

For the paragraph below, referring to the examples shown in the lecture slides, do the following by hand:

g) Count unigrams, bigrams, and trigrams and create a frequency list of each

('who', 'had', 'nearly') 2  
(The', 'wise', 'Sir Bedemir') 1  
(wise', 'Sir Bedemir', 'was') 1  
(Sir Bedemir', 'was', 'the') 1  
(was', 'the', 'first') 1  
(the', 'first', 'to') 1  
(first', 'to', 'join') 1  
(to', 'join', 'King Arthur') 1  
(join', 'King Arthur', 's') 1  
(King Arthur', 's', 'knights') 1  
(s', 'knights', ',') 1  
(knights', ', ', 'but') 1  
(', ', 'but', 'other') 1  
(but', 'other', 'illustrious') 1  
(other', 'illustrious', 'names') 1  
(illustrious', 'names', 'were') 1  
(names', 'were', 'soon') 1  
(were', 'soon', 'to') 1  
(soon', 'to', 'follow') 1  
(to', 'follow', ',') 1  
(follow', ', ', 'Sir Lancelot the Brave') 1  
(', ', 'Sir Lancelot the Brave', ',') 1  
(Sir Lancelot the Brave', ', ', 'Sir Galahad the

Pure') 1  
(', ', 'Sir Galahad the Pure', ',') 1  
(Sir Galahad the Pure', ', ', 'and') 1  
(', ', 'and', 'Sir Robin the Not-quite-so-brave-as-Sir-Lancelot') 1  
(and', 'Sir Robin the Not-quite-so-brave-as-Sir-Lancelot', 'who') 1  
(Sir Robin the Not-quite-so-brave-as-Sir-Lancelot', 'who', 'had') 1  
(had', 'nearly', 'fought') 1  
(nearly', 'fought', 'the') 1  
(fought', 'the', 'Dragon of Angnor') 1  
(the', 'Dragon of Angnor', ',') 1  
(Dragon of Angnor', ', ', 'who') 1  
(', ', 'who', 'had') 1  
(had', 'nearly', 'stood') 1  
(nearly', 'stood', 'up') 1  
(stood', 'up', 'to') 1  
(up', 'to', 'the') 1  
(to', 'the', 'vicious') 1  
(the', 'vicious', 'Chicken of Bristol') 1  
(vicious', 'Chicken of Bristol', 'and') 1  
(Chicken of Bristol', 'and', 'who') 1

(and', 'who', 'had') 1  
(who', 'had', 'personally') 1  
(had', 'personally', 'wet') 1  
(personally', 'wet', 'himself') 1  
(wet', 'himself', 'at') 1  
(himself', 'at', 'the') 1  
(at', 'the', 'Battle of Badon Hill') 1  
(the', 'Battle of Badon Hill', ',') 1  
(Battle of Badon Hill', ', ', 'and') 1  
(', ', 'and', 'the') 1  
(and', 'the', 'aptly') 1  
(the', 'aptly', 'named') 1  
(aptly', 'named', 'Sir Not-appearing-in-this-film') 1  
(named', 'Sir Not-appearing-in-this-film', ',') 1  
(Sir Not-appearing-in-this-film', ', ', 'Together') 1  
(', ', 'Together', 'they') 1  
(Together', 'they', 'formed') 1  
(they', 'formed', 'a') 1  
(formed', 'a', 'band') 1  
(a', 'band', 'whose') 1  
(band', 'whose', 'names') 1  
(whose', 'names', 'and') 1

(names', 'and', 'deeds') 1  
(and', 'deeds', 'were') 1  
(deeds', 'were', 'to') 1  
(were', 'to', 'be') 1  
(to', 'be', 'retold') 1  
(be', 'retold', 'throughout') 1  
(retold', 'throughout', 'the') 1  
(throughout', 'the', 'centuries') 1  
(the', 'centuries', ',') 1  
(centuries', ', ', 'the') 1  
(', ', 'the', 'Knights of the Round Table') 1  
(the', 'Knights of the Round Table', ',') 1

## REFERENCED:

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- Universal Dependencies: <https://universaldependencies.org>
- Daniel Jurafsky and James H. Martin. 2026. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition with Language Models*, 3rd edition. Online manuscript released January 6, 2026. <https://web.stanford.edu/~jurafsky/slp3>.

**Questions, remarks, pointers?**